Chaitanya Surabattuni

ML/AI Engineer | GenAI & Agentic AI Evaluation | Bay Area | LLM Systems
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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and evaluating GenAI and Agentic AI systems across fintech and healthcare. Currently building LLM evaluation infrastructure for an AI health coaching agent at PrimeHealth Technologies — designing ground-truth datasets, per-field accuracy metrics, LLM-as-judge frameworks, and CI-integrated regression suites. Prior production experience spans LangGraph agentic workflows, RAG pipelines, and ML systems at scale. Proficient in PyTorch, TensorFlow, HuggingFace Transformers, Python, and cloud-native MLOps (AWS, Docker, Kubernetes). AWS Certified Solutions Architect; MS in Artificial Intelligence (GPA 3.8), University of North Texas.

WORK EXPERIENCE

Jr. Agent Evaluation Engineer | **PrimeHealth Technologies** – Bay Area, CA *Apr 2026 – Present*

- Built ground-truth evaluation datasets of 60+ labeled examples across 21 health intent domains and adversarial edge cases to benchmark a multi-channel AI health coaching agent (chat, WhatsApp, voice); established per-field accuracy metrics across 5 classification dimensions driving domain routing accuracy from 71% to 93%.
- Designed and deployed an LLM-as-judge evaluation framework synthesizing 128 user personas and scoring AI-generated health recommendations on a structured 0-100 rubric, enabling scalable qualitative benchmarking at 15x the throughput of manual annotation.
- Identified and resolved a critical silent regression in the automated test suite where all classifier assertions passed vacuously, restoring meaningful CI signal and preventing undetected model drift across future model updates.
- Decoupled LLM-dependent regression tests from deterministic unit tests in CI, reducing average pipeline runtime by 60% while maintaining full coverage on non-LLM components.
- Defined KPIs and confusion matrix reporting for a 6-field structured output classifier; iterated on prompt design and output schema in collaboration with the model team to close failure mode gaps identified during evaluation.

Gen-AI Engineer – LLM Systems & Intelligent Automation | **LMES** – Dallas, TX *Jul 2025 – Apr 2026*

- Designed and deployed production LLM-powered voice and chat agents using Python and LangGraph; applied advanced prompt engineering to maintain brand consistency and eliminate hallucinations across customer-facing channels.
- Built RAG-based information retrieval pipelines integrating structured (SQL/DynamoDB) and unstructured data sources, enabling automated resolution of complex member queries across 10+ microservices.
- Achieved up to 90% reduction in model-serving costs via hybrid local/cloud execution, Redis-based semantic caching, and async architectures; implemented CI/CD via GitHub Actions and Docker enabling rollback-ready releases.
- Conducted continuous QA on AI agent conversations — analyzing logs, identifying failure modes, and iterating on prompts and decision logic to improve resolution rates and interaction quality.
- Implemented TDD and unit testing frameworks for AI microservices, improving reliability and enabling safe iterative model updates across concurrent AI workloads.

AI/ML Engineer Intern – Fintech Intelligence | **3 Pillars Equity** – Frisco, TX *Jun 2024 – Aug 2024*

- Designed AI-powered anomaly detection models integrated into financial transaction pipelines, flagging suspicious activity in real time and reducing false positives by 30%.
- Developed an LLM-backed text-to-SQL natural language query interface (Python, LangChain) over financial databases, reducing analyst overhead by 25% and enabling non-technical stakeholders to extract insights autonomously.
- Built and productionized an ML-based credit risk scoring system replacing manual rule-based logic, improving loan decision throughput by 40% while maintaining full regulatory compliance.
- Containerized all AI services with Docker and automated CI/CD workflows; mentored junior developers on OOP and clean architecture principles.

Data Scientist – Analytics & ML Modeling | **GD Research Pvt Ltd** India *May 2022 – Jul 2023*

- Developed and deployed predictive ML models to support enterprise decision-making; improved downstream analytics accuracy by 35% across multiple business data streams through feature engineering and model optimization.
- Built NLP-based entity extraction and text classification pipelines to transform unstructured data into structured insights, enabling scalable analysis across large enterprise knowledge bases.
- Designed and automated end-to-end data ingestion and processing workflows using Apache Kafka and vector embeddings; enabled semantic search and faster data discovery for analytics teams.
- Collaborated with engineering teams to productionize data pipelines and models; established coding best practices and mentored junior analysts on data modeling and OOP standards.

Machine Learning Intern | **EdGate Technology** – Bengaluru, India *Aug 2020 – Dec 2020*

- Architected production NLP conversational AI chatbot (Python, TensorFlow, NLTK) with intent classification and entity extraction, improving automated query resolution by 30%.
- Built a computer vision pipeline using CNNs achieving 95% image recognition accuracy; executed full ML lifecycle from data cleaning through production model integration.

KEY PROJECTS

Multi-Agent AI Automation Platform | *LangGraph, Python, RAG, SQLite, Docker*

- Architected a LangGraph-based agentic workflow orchestrating 3 specialized AI agents (Researcher / Coder / Reviewer) with human-in-the-loop checkpoints, iterative review cycles, and SQLite persistent session state; implemented TDD and modular OOP design enabling clean agent extension and unit testing.
- Implemented RAG pipeline over structured and unstructured data — directly analogous to Qualcomm's on-device GenAI and embedded RAG use cases requiring efficient retrieval with minimal resource overhead.

High-Throughput ML Inference Microservice | *NVIDIA Triton, Kubernetes, Kafka, Redis, Prometheus/Grafana*

- Engineered a production GPU inference service on Kubernetes using NVIDIA Triton with async batching via Kafka and Redis, achieving 40% P99 latency reduction; deployed Prometheus/Grafana observability stack with real-time SLA alerting.
- Experience with GPU/accelerator-optimized inference directly applicable to Qualcomm NPU/GPU acceleration use cases.

Financial Churn & Risk Prediction System | *Python, Scikit-learn, Pandas, MySQL, SQL*

- Designed a supervised ML pipeline predicting member churn and LTV using classification and regression models; built evaluation framework tracking accuracy, precision/recall, and ROC-AUC with dashboards surfacing risk segments.

TECHNICAL SKILLS

ML & AI Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, NVIDIA Triton, TorchServe, OpenCV, Scikit-learn
GenAI / LLM / Agentic AI: LangGraph, LangChain, Prompt Engineering, RAG, Vector Databases (Qdrant, Chroma), Multi-Agent Orchestration, Tool Use / Function Calling
Agent Evaluation & QA: LangSmith, Benchmarking Pipelines, KPI Design (F1/ROUGE/Hallucination Rate/Latency), Automated Test Harnesses, Failure Mode Analysis, TDD
NLP & Conversational AI: HuggingFace, NLTK, Intent Classification, Voice/Chat Agent Development
Inference Optimization: NVIDIA Triton, Async Batching, Redis Semantic Caching, Hybrid Local/Cloud Execution, Kubernetes HPA
Certifications: AWS Certified Solutions Architect – Associate, MLOps

EDUCATION

MS – Artificial Intelligence | University of North Texas, Denton, TX | GPA: 3.8 *Aug 2023 – May 2025*
BE – Electronics & Communication Engineering | Visvesvaraya Technological University, India | GPA: 3.7 *Aug 2018 - May 2022*